

Worksheet · Section 3.3–3.4

Mathematical Expressions & Strings



The Big Picture

Algorithms are built from sequencing, selection, and iteration. Practice MOD, order of operations, and string procedures like concatenation and substring.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|------------------------------------|--|
| ____ 1. Expression | A. An ordered sequence of characters |
| ____ 2. Sequencing | B. Gives the remainder after division |
| ____ 3. Selection | C. The rules for which operation happens first |
| ____ 4. Iteration | D. Repeating steps |
| ____ 5. MOD | E. A finite set of instructions that accomplish a task |
| ____ 6. Order of operations | F. Joining strings end-to-end |
| ____ 7. String | G. A part of an existing string |
| ____ 8. Concatenation | H. Choosing which steps run based on a condition |
| ____ 9. Substring | I. A combination of values and operators that evaluates to a single value |
| ____ 10. Algorithm | J. Running steps in the order they are written |

Part 2 · Fill It In

Use each word from the word bank once.

sequencing

even

substring

0

concatenation

parentheses

characters

remainder

1. $a \text{ MOD } b$ gives the _____ when a is divided by b .
2. If $n \text{ MOD } 2 = 0$, then n is _____.
3. $15 \text{ MOD } 5 =$ _____.
4. Running steps in the order they are written is _____.
5. Joining “base” and “ball” into “baseball” is _____.
6. “port” is a _____ of “important”.
7. A string is an ordered sequence of _____.
8. To make the addition happen first in $2 + 3 * 4$, use _____.

Name: _____

Class: _____

Date: _____

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Practice

Part 3 · MOD Drill

Expression	Value
14 MOD 4	_____
20 MOD 6	_____
7 MOD 7	_____
3 MOD 9	_____
100 MOD 10	_____
58 MOD 10	_____
(9 + 7) MOD 5	_____
9 + 7 MOD 5	_____

Part 4 · Order of Operations

Expression	Value
$2 + 3 * 4$	_____
$(2 + 3) * 4$	_____
$18 / 3 * 2$	_____
$10 - 4 \text{ MOD } 3$	_____
$6 * 2 \text{ MOD } 5$	_____

MOD has the same precedence as _____, so work left to right.

Part 5 · String Lab

`substring(str, start, length)` returns *length* characters starting at *start* (first position is 1). `word ← "COMPUTER"`

- `substring(word, 1, 4) →` _____
- `substring(word, 5, 4) →` _____
- `concat(substring(word, 4, 3), "S") →` _____
- Number of characters in word → _____
- `concat("AP", concat(" ", "CSP")) →` _____

Part 6 · MOD in Action

- An expression that is true when *n* is a multiple of 3: _____
- A clock shows hour *h* (0–23). The hour 5 hours later: _____
- A statement that displays the last digit of *n*: _____
- Every 4th customer gets a coupon: IF (_____)



Think:

MOD = what's left over. 23 MOD 4: 4 goes into 23 five times (20) with 3 left. And spaces count as characters in strings!

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. What is $23 \bmod 5$?

- A. 5
- B. 4
- C. 4.6
- D. 3

★★ On Level

2. Which expression is true for every odd positive integer n ?

- A. $n \bmod 2 = 1$
- B. $n \bmod 1 = 1$
- C. $n / 2 = 1$
- D. $n \bmod 2 = 0$

★★ On Level

3. Using the substring definition above, what is `concat(substring("TEACHER", 1, 3), "M")`?

- A. "TEACM"
- B. "TEAM"
- C. "TEA M"
- D. "MTEA"

★★★ Exam Hard

4. What are x and y ?

```
x ← 17
y ← x MOD 5
x ← (x - y) / 5
```

- A. $x = 17, y = 2$
- B. $x = 3, y = 2$
- C. $x = 2, y = 3$
- D. $x = 3.4, y = 2$

★★★ Exam Hard

5. For a positive integer n , which TWO expressions always give the ones digit? (*Select two.*)

- A. $n \bmod 10$
- B. $10 \bmod n$
- C. $n / 10$
- D. $(n \bmod 100) \bmod 10$

Part 8 · Written Response

A cafe gives a free drink to every 10th customer. Write the condition using MOD, and explain how it works for customers 9, 10, and 20.
